COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212-Automating Server Management	**SECTION:**	CPE31S2 -
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Hands-on Activity 8.1 Install, Configure, and Manage Enterprise Availability Monitoring via Ansible

1. PROCEDURE OUTPUT

Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating Roles.

To create an ansible playbook we need to create the inventory first this is the remote server where nagios will be installed.

Inventory.yml:

```
all:
  children:
    nagios-ubuntu:
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.102
          ansible_user: server-1
    nagios-centos:
      hosts:
        centos1:
          ansible_host: 192.168.56.106
          ansible_user: cent-server

    nagios-server:
      hosts:
        ubuntu1:
        centos1:
```

Check the list for confirmation if the script was implemented correctly

Command used:
Ansible-inventory -i inventory.yml --list

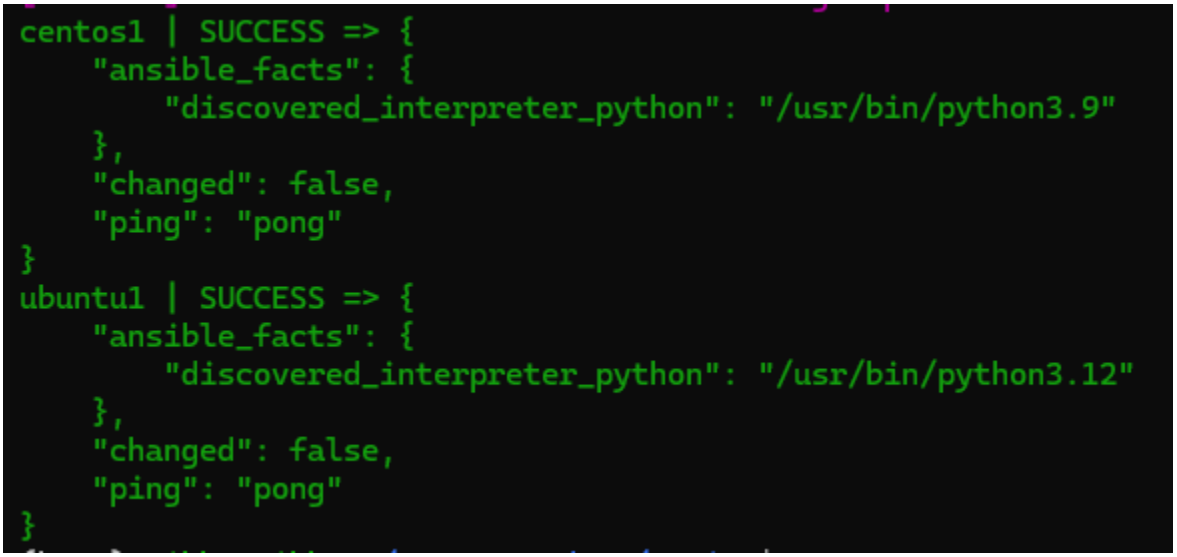
Output:



If the script is running successfully the last step is to check if the connectivity via ssh is working

Command used:
Ansible all -i inventory -m ping

Output:



Now that the inventory and the connectivity is working the next step is to create the cfg file for ansible so that we don't have to define inventory file when running the playbook

Ansible.cfg:

```
[defaults]

inventory = inventory.yml
roles_path = roles
host_key_checking = False
retry_files = False
interpreter_python = auto_silent
```

This is just a simple way to simplify the running command

Example command:

Ansible-inventory --list & ansible all -m ping

Next step would be creating the site.yml, site is the orchestrator of the entire structure, this basically consolidates the roles configuration and plays that the users want to execute. We can think of it as the structured folder.

Site.yml:

```
---
- name: install magios core
  hosts: nagios-server
  become: true
  gather_facts: true

  roles:
    - role: nagios
```

After this the next step is to add a roles directory this will be the main structured folder where the changes in our remote servers will be controlled.

Add a handlers folder for starting/restarting/stopping a service inside of that folder will be the main.yml where the script lived:

```
---
- name: Restart apache on ubuntu
```

```
  ansible.builtin.service:
    name: apache2
    state: restarted
  listen: restart web server
  when: ansible_distribution == "Ubuntu"

- name: Restart httpd on centos
  ansible.builtin.service:
    name: httpd
    state: restarted
  listen: restart web server
  when: ansible_distribution == "CentOS"

- name: Restart Nagios
  ansible.builtin.service:
    name: nagios
    state:  restarted
  listen: restart nagios
```

Next would be to add a tasks/main.yml the tasks folder is the folder where the installation will be downloaded in our remote:

```
---

- name: Update package cache on Ubuntu and CentOS
  ansible.builtin.package:
    update_cache: true

- name: Install nagios in ubuntu
  ansible.builtin.apt:
    name:
      - nagios4
      - nagios-plugins-basic
    state: present
    update_cache: yes
    when: ansible_distribution == "Ubuntu"
    notify: restart nagios ubuntu

- name: Install EPEL repo on CentOS
```

```
  ansible.builtin.dnf:
    name: epel-release
    state: present
  when: ansible_distribution == "CentOS"

- name: Install nagios on centos
  ansible.builtin.dnf:
    name:
      - nagios
      - nagios-plugins-all
    state: present
    when: ansible_distribution == "CentOS"
    notify: restart nagios centos

- name: Ensure that nagios is ruuning and enabled
  ansible.builtin.service:
        name: "{{ 'nagios4' if ansible_distribution == 'Ubuntu' else
'nagios4' }}"
    state: started
    enabled: yes
```

Expected output when playbook is executed:

Currently updating both packages on centos and ubuntu:

```
(base) nikko@Nikko:~/CPE212_Andoy-/nagios$ ansible-playbook -bK site.yml
BECOME password:

PLAY [Install and configure Nagios Core] ***************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [centos1]
ok: [ubuntu1]

TASK [roles : Update package cache on Ubuntu and CentOS] ***********************************************************
changed: [ubuntu1]
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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